## Contact

fabzy4l@gmail.com

www.linkedin.com/in/fpalvarez23 (LinkedIn)


## Top Skills

Presentation Skills
Data Analytics
Characterization

## Languages

French
Spanish
Japanese
Latin

## Certifications

Vaccine Development
MATLAB Fundamentals
Automotive Electronics
X-Ray Radiation Safety

## Honors-Awards

The General Property Scholarship for Graduates
Eloise & Patrick Wieland Fellowship
Dissertation Completion/Excellence Fellowship
NSF-PREM Scholar
MRS Scholar/presenter

## Publications

Fabrication of surfactant-dispersed HiPco single-walled carbon nanotube-based alginate hydrogel composites as cellular products.
Alginate Hydrogels with Embedded ZnO Nanoparticles for Wound Healing Therapy
Failure Analysis of Additively Manufactured Polyester Test Specimens Exposed to Various Liquid Media

# Fabian Alvarez-Primo

Bioengineering, Stem Cells, Materials Science, Data & Statistics
El Paso, Texas, United States

## Summary


I love constant improvement.

CliftonStrenghts: Achiever, Learner, Input, Ideation, Intellection.

Always looking for opportunities to grow.

I am a proud Mexican-American scientist/engineer & entrepreneur with very strong technical foundations, holding a Ph.D. in Materials Science Engineering primarily focused on Biomedical Engineering and Stem Cell based Regenerative Medicine from a highly competitive R1 institution: The University of Texas - El Paso.

Professionally, I have become proficient in the management and application of resources to resolve materials engineering, biomedical and molecular biology problems and develop biotechnological applications in clinical and non-clinical settings.

Further, I have >10 years of entrepreneurial and sales skills, having demonstrated the ability to build and promote small businesses across several industries, having also helped in the development of Bioengineering/Biotech ventures, where I have also applied my experience.

My areas of experience are:

- Materials Science & Engineering
- Regenerative Med / Tissue Engineering
- Neuroscience / Neuropsychiatric Research
- Cardiovascular Research
- Stem Cell Research
- Data Analysis / Statistics / Bioinformatics
- Molecular Biology / Genomics
- Physical, Analytical & Clinical Chemistry
- Vaccine Design / Drug Delivery / Biosensors

A comparative study in the printability of a bioink and 3D models across two bioprinting platforms.

Bone tissue engineering techniques, advances and scaffolds for treatment of Bone Defects.

- 3D printing / bio-printing / CAD modeling
- Electronics, Mechanical Design & Mechanical Testing of biomaterials, inorganics and devices

Hobbies:

• I am an evidence based powerlifter with 5+ years of experience, a martial artist and an athlete.

• My preferred reads are in Finance, Personal Development, Bioengineering, Neuroscience, Excercise, Genetics and anything that merges business development, computers, mechanics, engineering with biology, chemistry and medicine.

## Experience

### Texas Tech University Health Sciences Center El Paso

Neuropsychiatry Post Doctoral Fellow

May 2022 - November 2022 (7 months)

El Paso, Texas, United States

• In charge of a project consisting of reprogramming human blood cells (PBMCs) via non-integrating methods, into clinically relevant induced Pluripotent Stem Cells compliant per cGMP standards for downstream applications, including induction into other cell types for pharmacodynamic modeling.

• In charge of data structuring, management and conducting biostatistical data analysis for several research projects. (link available below).

• Conducted biobanking, dissections, processing, and measuring of biomarkers in postmortem human brain tissue, in association with the Southwest Brain Bank and El Paso County Coroner’s Office.

• Conducted molecular biology testing methodology (as listed in skills & interests), and characterization of the pathophysiology of neuropsychiatric disorders through bioinformatics: genomic and transcriptomic sequencing via RNA-Seq.

### West Texas Pain Institute, PLLC

Toxicology & Pharmacogenomics Clinical Laboratory Scientist/Manager

March 2021 - April 2022 (1 year 2 months)

El Paso, Texas, United States

• Supervised Clinical Laboratory Testing of ~1300+ samples monthly and Laboratory Information Management System (LIMS - StratusDX) within a Pain Medicine practice, successfully leading a team of 9 people and performing administrative duties in synergy with healthcare providers to ensure quality pharmacological monitoring and patient care, while successfully maintaining CLIA, COLA, and HIPAA regulatory adherence.

• Improved workflow, patient pharmacological, and documentation monitoring of EMR information (eClinicalWorks) through efficient LIMS data management, processing, analysis, and on-the-go application of solutions obtained through analytical methods conducted via data Excel, R, and Python.

• In charge of the supervision, operation through DHHS guidelines and Drug of Abuse protocols, maintenance and/or troubleshooting of characterization equipment such as Clinical Biochemistry Analyzer, Enzyme Immunoassays (CEDIA, DRI, QMS, EIA) and Liquid Chromatography/Mass Spectrometry for confirmatory analysis. Supervised DNA laboratory and Pharmacogenomic testing methodology compliance.

### Radford School

Teacher

January 2020 - December 2020 (1 year)

El Paso, Texas, United States

High School teacher of science disciplines, including:

• Introductory level Electromagnetics, Biology, Molecular Biology, Kinetics, Electronics, Classical Mechanics, Introductory Quantum Mechanics, Environmental Science, Intro to Nuclear Physics, General Chemistry, Fluid Mechanics, Physical Chemistry, etc.

### Biosvelte

Chief Scientific Officer/Co-Founder

January 2018 - December 2020 (3 years)

United States

• Cofounded, designed, researched and managed a team of 10 people in charge of sales for an ‘evidence based in-house designed health supplement products’ to promote weight loss, fitness and metabolic health.

• In charge of marketing strategies, optimized through Data mining, management & analytics.

• As active CSO, I was in charge of all the chemistry-associated quality control aspects of the product development, fabrication and distribution.

• Sales.

### ProScribe
Medical Scribe
August 2019 - October 2020 (1 year 3 months)
El Paso, Texas, United States

• Management of clinical documentation, through electronic medical records (eCW - eClinicalWorks), in assistance to healthcare providers within a variety of clinics.

• Monitoring entry of CBC, CMP, PT, BMP, LP, HA1C, UA, Cultures, TSH, and other screening procedures, as well as updating the history of present illness, patient history, physical exam, treatment, diagnosis, and other pertinent health information in the patient record.

### Inspired Materials & Stem cell based Tissue Engineering Lab (IMSTEL)
Doctoral Research Assistant
August 2016 - July 2020 (4 years)
El Paso, Texas

• Responsible for the development and characterization of a conductive cardiovascular engineered tissue composite, as a vehicle for stem cell therapy and other cellular products. Methodology compliant with current ASTM and ISO 9001 standards. Work in collaboration with Marti’s Lab at RICE University.

• Contributed to development, design and/or testing of: (1) antibacterial composites for diabetic wound healing, (2) engineered bone, (3) skin printing, (4) a mathematical model of synthetic tissue degradation in physiologic conditions, (5) protein-based drug delivery vehicle for neglected tropical disease therapeutics, and (6) biosensors.

• In charge of data management and biostatistical data analyses on all projects.

• Mentor 30 graduate/undergraduate students on the use of Scanning Electron Microscopy and 3D Bioprinting for the MMBME department to maximize student learning. Also taught Microscopy portion of the class entitled “BME 4371 - Biomaterials Characterization and Biocompatibility" to undergraduate and graduate students, resulting in a publication.

### Alorica
Customer Service Sales Representative
October 2016 - February 2018 (1 year 5 months)
El Paso, Texas, United States

An opportunity to practice my people skills and improve my ability to sell.

• Commission based sales representative for DirecTV/AT&T. Job required maintaining an "SPH (sales per hour) goal". Customer service oriented telemarketing was also a crucial part of the duty as an agent.

• Promoted to Trainer, due to good performance and recommendations.

• In charge of training classes of up to 20 people each, teaching them how to sell and handle emotional intelligence based interactions with upset customers.

• Taught about 8 classes, for a total of 160 people trained. Dialed and gave satisfactory performances in all the projects available to the company.

• Employee of the Month for April 2017. Received "Starz performance award" for best performance.

### New Financial Solutions, LLC
Chief Executive Officer/Co-founder
January 2016 - December 2017 (2 years)
El Paso, Texas, United States

• Directed, analyzed and formulated real estate market strategies aimed at successful property acquisition and resale.

• Obtained real estate leads from privately owned companies for the metropolitan areas of El Paso and San Antonio, and conducted financial data analytics to maximize profitability.

• Managed a team of 6 people, alonside a board of investors acting as a committee to assess viability of property acquisition.

• Ensured customer service excellence.

• Facilitated resource availability for customers, to assist in smooth transitions.

### Mathmobile Tutorials
Interdisciplinary Tutor
August 2015 - January 2016 (6 months)
El Paso, Texas, United States

• Tutoring of subjects such as Biology, Mathematics, Algebra, Calculus, Pre-Calculus, Physics and Chemistry.

### Climax Molybdenum
R&D Engineer Intern
May 2015 - August 2015 (4 months)
Sahuarita, Arizona

• In-charge of project regarding materials characterization via analytical chemistry and mechanical testing of compressive strength for a refractory ceramic material.

• Lead the improvement initiative on endurance of the material and life-expectancy, intended for practical processing methods in the industry (Leaching Vessels).

• Conducted analytical chemistry techniques to determine density analysis and weight loss analysis in correlation to the curing factors and their effects on the material.

• Testing conducted involved vast knowledge of standards and regulations complying to the ASTM and statistical analysis to successfully develop a Standard Operating Procedure on proper material handling.

### W.M. Keck Center for 3D Innovation
Undergraduate Research Assistant
November 2014 - May 2015 (7 months)

The University of Texas at El Paso

• Worked on projects that consisted in the synthesis and testing of mechanical properties (Tensile, compression, creep) of polymer based composite materials which were also extruded into filaments and turned into 3D printed products by us.

• Characterized through XRD and SEM crystal structures and fracture surfaces of ASTM D638 samples, respectively.

• Improved materials by modifying constituent ratios and intrinsic properties, tailoring them to their specific applications, ranging from Biomedical (PLA) to Aerospacial (ABS/SEBS).

• Injection molded polymeric ASTM D638 samples for comparison to 3D printed samples and independently conducted analysis of anisotropic vs. isotropic properties as a result from the two processing methods.

• Contributed on the Wohlers Report 2015: Additive Manufacturing and 3D Printing State of the Industry Annual Worldwide Progress Report.

### Materials Research & Technology Institute
Undergraduate Research Assistant
June 2014 - August 2014 (3 months)
El Paso, Texas

Assisted in writing and researching in two different projects consisting of organic chemistry and materials science, in collaboration with Universidad Metropolitana Ana G Mendez, based in Puerto Rico. Projects were as follows:

• Bismuth Nanoparticles' synthesis, characterization and application as semi-conducting materials, based on effective mass.

• Cobalt/Nickel Molydbenum Sulfides based catalyst for toxin chelation & purification of aqueous media in compliance with EDA regulations. (worked under Dr. Russell Chianelli, Stanford’s Top 2% Scientist ranking)

Assisted in the paper, "Bismuth nanoparticles: A new route of synthesis and their morphological aspects," for a high-profile publication.

### LIMBS International
Undergraduate Research Assistant

January 2012 - August 2012 (8 months)
United States

• Assisted in design of prosthetics, focused primarily on researching a variety of materials for their applicability in the manufacturing of non-expensive prosthetics for children in 3rd world countries that do not count with the funds/ resources to afford it. Under former Director of LIMBS, Dr. Roger Gonzalez.

## Education

### The University of Texas at El Paso

Doctor of Philosophy - PhD, Biomedical Materials Science & Engineering · (2017 - 2020)

### The University of Texas at El Paso

Bachelor's degree, Metallurgy & Materials Engineering · (2010 - 2015)